

Ankush

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SUMMARY

Computer Science undergraduate skilled in Java, full-stack development, and cloud technologies, with internship experience building AI-assisted applications. Developed end-to-end projects using React, Node.js, PostgreSQL, MongoDB, and REST APIs, including a Java-based Order Matching Engine. Hackathon winner with experience in secure and scalable software systems.

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering — CGPA: 8.81

Vellore, Tamil Nadu

EXPERIENCE

Lampros Tech Labs Pvt. Ltd.

May 2026 – July 2026

- Developed features for an AI-assisted ride dispatch and tracking platform supporting ride booking, driver allocation, and route optimization.
- Implemented secure authentication, intelligent ride allocation, and Google Maps integration for real-time tracking and navigation.
- Contributed to the development of scalable backend workflows for ride dispatch and management.

PROJECTS

BookBase – Book Review System | *React.js, Node.js, Express.js, MongoDB, JWT*



- Developed a full-stack digital library and book review platform with JWT-based authentication, enabling users to discover, rate, review, and manage books through a responsive interface.
- Integrated Google Books API for real-time book discovery and YouTube Data API to provide dynamic audiobook content within the application.
- Built personalized features including My Library, favourites, category-based search, notifications, persistent reviews, and RESTful MongoDB-backed data management.

Order Matching Engine | *Java, JUnit 5, Maven*



- Designed and implemented a limit order-matching engine that replicates core stock exchange functionality using price-time priority and FIFO order execution.
- Engineered an efficient order book using **TreeMap**-based price levels and doubly-linked lists, enabling $O(\log n)$ price lookup and $O(1)$ order cancellation.
- Implemented partial fills and multi-level order matching, validated through a 12-case JUnit 5 test suite covering matching, cancellation, and edge cases.

Smart Ride Dispatch System | *React.js, Tailwind CSS, Node.js, Express.js, PostgreSQL*



- Developed an AI-assisted ride dispatch platform that intelligently matches riders with suitable drivers using proximity, estimated travel time, and availability-based scoring.
- Integrated Google Maps API for location selection, route visualization, distance calculation, and traffic-aware ETA to improve ride assignment and navigation.
- Built secure role-based authentication, RESTful APIs, and PostgreSQL-backed workflows supporting ride booking, dispatch, real-time tracking, and completion.

CERTIFICATIONS

AWS Solutions Architect (SAA-C03)

Oracle Cloud Infrastructure AI Foundations



TECHNICAL SKILLS

Languages: Java, JavaScript, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, REST APIs

Databases: MongoDB, MySQL

Cloud: AWS (EC2, S3, IAM, VPC, RDS)

Tools: Git, GitHub

RESPONSIBILITIES & ACHIEVEMENTS

Hackathon – CITADEL Secure Military Communication System

- Built a tamper-resistant battlefield communication framework with AES-256 + ECC hybrid encryption, blockchain-based validation, and ESP32 hardware integration to prevent replay and tampering attacks.